

Alessandro Pasqui

Curriculum Vitae

17, Avenue Stephen Pichon
75013 Paris, France
+33 629644877
✉ ap.pasqui@gmail.com
🌐 apasqui.github.io



Personal Information

Date of birth 29 - 11 - 1995
Place of birth Rome, Italy

Education

Sep 2021 – Oct 2025 (expected) **MSCA PhD Candidate at PSL Université de Paris** 
**Collège de France, Centre Interdisciplinaire de Recherche en Biologie (CIRB)
& École Normale Supérieure**

Subject: AI-based Methods for Inverse Problems and Computer Vision

Supervisor: Dr. Hervé Turlier

AI-based methods for inverse problems in biological cell systems

In the first phase of the PhD, developed *ZAugNet*, a self-supervised, machine learning-based method for z-slice augmentation in 3D bio-imaging. This approach integrates concepts from adversarial learning and knowledge distillation. The manuscript is available here: <https://arxiv.org/abs/2503.04843>.

In the second phase, developed *VertAX*, a computational framework for efficient vertex-based modeling and inverse problem solving via bilevel optimisation in confluent tissues. Implemented in JAX, *VertAX* is fully differentiable through automatic differentiation and allows the inference of optimal cellular parameters to reproduce a desired macroscopic behavior or to fit experimental data. A manuscript on this work is currently in preparation.

Sep 2018 – Mar 2021 **Master's Degree at Sapienza University of Rome** 
**Department of Physics
& Italian Institute of Technology, Center For Life Nano Science (CLNS-IIT)**

Subject: Theoretical Physics (Statistical Mechanics)

Supervisors: Prof. Giancarlo Ruocco, Dr. Giorgio Gosti, Dr. Mattia Miotto

High performance matching algorithm based on orthogonal polynomials

Developed a computational algorithm to detect complementary-shaped object portions, applied to 2D jigsaw puzzles and 3D archaeological artifacts. The method used local shape descriptors based on orthogonal polynomials. This work contributed to a broader CLNS-IIT project on spike protein–receptor interactions in viral infections (Milanetti et al. 2020).

Grade: 110 (with honors) / 110

Sep 2014 – Jan 2018 **Bachelor's Degree at Sapienza University of Rome** 
Department of Physics

Subject: Physics

Supervisor: Prof. Federico Ricci-Tersenghi

Percolation in models with long-range interactions

Studied, through theoretical analysis and numerical simulations, the critical point and critical exponents in percolation models involving long-range interactions.

Research Experiences

- Sep 2023 – Dec 2023 **Italian Institute of Technology, Center for Life Nano Science (CLNS-IIT)**
Rome, Italy
Visiting Researcher
Supervisors: Prof. Giancarlo Ruocco, Dr. Giorgio Gosti, Dr. Mattia Miotto
Visiting period as part of a collaboration to test VertAX, a differentiable vertex-model that I had developed to solve inverse problems on real microscopic data for confluent tissues.
- Apr 2021 – Aug 2021 **Collège de France, Centre Interdisciplinaire de Recherche en Biologie**
Paris, France
Research Engineer
Supervisors: Dr. Hervé Turlier
Image-to-image translation via convolutional neural networks for microscopy images in the context of developmental biology: from transmitted light microscopy to fluorescence microscopy (and back).
- Jan 2020 – Feb 2020 **Sapienza University of Rome**
Rome, Italy
Intern
Supervisor: Dr. Francesca Colaïori
Connection properties of the directed graph related to the election results for new Wikipedia administrators.
- Mar 2019 – Jul 2019 **Italian Institute of Technology, Center for Life Nano Science (CLNS-IIT)**
Rome, Italy
Intern
Supervisors: Prof. Giancarlo Ruocco and Dr. Giorgio Gosti
Pairwise correlation function for neural cells during neural rosettes development through automatic segmentation algorithms based on deep learning using convolutional neural networks.

Programming

JAX/Pytorch	excellent
Python	excellent
LaTeX	excellent
C/C++	advanced
Shell Scripting	advanced

Languages

English	fluent
French	fluent
Italian	fluent
Spanish	advanced

Honors and Awards

- Jun 2024 **II EELISA Student Scientific Competition**
Budapest, Hungary
Awarded *first prize* for outstanding research contribution in the Chemical and Bioengineering category. Featured in multiple interviews highlighting the scientific and societal impact of the project: [EELISA interview](#), [PSL interview](#).
- Aug 2021 **Marie Skłodowska-Curie Actions (MSCA) - COFUND Fellowship**
AI4theSciences Doctoral Program 2021 at PSL Université Paris
Paris (France)
Awarded a *Marie Skłodowska-Curie Actions (MSCA) COFUND Fellowship* as part of a highly competitive European PhD program, recognizing academic excellence and research potential.

List of Publications

- 2025 **Self-Supervised Z-Slice Augmentation for 3D Bio-Imaging via Knowledge Distillation**
Preprint available on arXiv: <https://arxiv.org/abs/2503.04843>
- 2025 (expected) **VertAX: Inverse Vertex-Based Modeling for Tissue Mechanics**
Differentiable computational framework for efficient vertex-based modeling and inverse problem solving via bilevel optimisation in confluent tissues. Manuscript in preparation.
- 2025 (expected) **AlphaMIC: Synthetic Microscopy Image Generation via CycleGANs**
In collaboration with the [Turlier Lab](#). Development of a software tool to generate realistic artificial microscopy images using cyclic generative adversarial networks. Manuscript in preparation.
- 2025 (expected) **3D Nuclear Roughness Estimation in Patient-Derived Organoids**
In collaboration with [Dr. Kate Miroshnikova](#). Analysis of 3D nuclear morphology in patient-derived colonic organoids using spherical harmonics as local shape descriptors for pathological condition classification. Manuscript in preparation.

Conferences and Talks

- Feb 2025 **GDR IASIS - Unrolling and un/self/supervised learning for inverse problems**
Inria Paris, France
Poster presentation.
- Feb 2025 **Les Houches Physics School - From Soft Matter to Biophysics**
École de physique des Houches, France
Poster presentation.
- Sep 2024 **Institute of Physics - AI for Biological Physics**
IoP London, United Kingdom
Talk (30 minutes).

Jul 2024 **World Congress on Computational Mechanics (WCCM/PANACM)**
Vancouver, Canada
Talk (20 minutes).

May 2024 **PSL Doctoral Conference - AI4theSciences**
PSL Paris, France
Talk (20 minutes).

Peer Review

Apr 2023 **21st IEEE International Symposium on Biomedical Imaging (ISBI)**
Athens, Greece
Served as a peer reviewer for the “Light My Cells: Bright Field to Fluorescence Imaging Challenge,” organized by the French National Research Infrastructure for Biological Imaging (FBI) as part of the ISBI 2024 conference.

Organizing Committees

Mar 2023 – Present **CIRB Seminar Series at Collège de France**
Paris, France
Member of the organizing committee for the weekly seminar series of the Biology Department at Collège de France. Responsibilities include inviting speakers, chairing sessions, and coordinating post-seminar receptions.

Other Experience

Jan 2016 – Dec 2016 **Librarian at Sapienza University of Rome, Department of Physics**
Rome, Italy
Selected for a merit-based internship to support library operations. Responsibilities included assisting patrons, managing check-in/check-out of materials, and classifying/cataloguing books and publications using standard library classification systems.

Volunteering

Oct 2024 **French Science Festival at Collège de France**
Paris, France
Organized and ran a public science outreach stand at the Collège de France as part of the French Science Festival, aimed at making scientific research accessible to a general audience.

Sep 2024 **European Heritage Days at Collège de France**
Paris, France
Contributed to the organization of events for the European Heritage Days at the Collège de France, helping welcome visitors and explain the historical and scientific significance of the institution.

Contacts for References

Dr. Hervé Turlier

Group Leader at Collège de France (CIRB)

Email: herve.turlier@college-de-france.fr